

Variational Principles

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How do we find the shortest path between two points – in Euclidean space, on the surface of a sphere? What about on a general curved surface?

Definition 0.1 (Functionals). A *functional* is a map from a function to $\mathbb{R}$.

The distance between a curve is a functional from the parameterisation of that curve to its distance. So we would like to find a way to determine minima and maxima of functionals.

§1 Calculus on $\mathbb{R}^n$

Consider a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, and assume that f is nice and differentiable (at least C^2).

Definition 1.1 (Stationary points). We say that $\mathbf{a} \in \mathbb{R}^n$ is a *stationary point* of f if $\nabla f(\mathbf{a}) = \mathbf{0}$.

Around such a point, we have a Taylor series given by

$$f(\mathbf{x}) = f(\mathbf{a}) + \underbrace{\sum_i (x_i - a_i) \nabla f(\mathbf{a})}_0 + \frac{1}{2} \sum_{i,j} (x_i - a_i)(x_j - a_j) H_{ij}(\mathbf{a}) + O(|\mathbf{x} - \mathbf{a}|^3)$$

where $H_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j}$ is the symmetric Hessian matrix. Let us assume without loss of generality that $\mathbf{a} = \mathbf{0}$, and we obtain

$$f(\mathbf{x}) - f(\mathbf{0}) = \frac{1}{2} x_i H_{ij} x_j + O(|\mathbf{x}|^3)$$

under the summation convention, where we are implicitly evaluating H at $\mathbf{0}$. Since H is symmetric, we may diagonalise with a special orthogonal/rotation matrix R satisfying $x_i = R_{ij} x'_j$ and

$$H' = R^\top H R = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix}$$

for real eigenvalues $\lambda_1, \dots, \lambda_n$. We now obtain

$$f(\mathbf{x}) - f(\mathbf{0}) = \frac{1}{2} \sum_{i=1}^n \lambda_i (x'_i)^2 + O(|\mathbf{x}|^3)$$

Hence, we conclude that

- if every $\lambda_i > 0$, then $f(\mathbf{x}) > f(\mathbf{0})$ for small $\mathbf{x}$ – we are at a local minimum;
- if every $\lambda_i < 0$, we are analogously at a local maximum;
- if there are mixed eigenvalues, we are at a saddle point.

However, we must worry about *degenerate* cases, where at least one $\lambda_i = 0$ – it is then necessary to go to higher-order terms of the Taylor expansion.

Let us consider specifically the $n = 2$ case, in which

$$\det H = \lambda_1 \lambda_2 \quad \text{Tr } H = \lambda_1 + \lambda_2$$

so that

- $\det H > 0, \text{Tr } H > 0 \implies$ local min;
- $\det H > 0, \text{Tr } H < 0 \implies$ local max;
- $\det H < 0 \implies$ saddle;
- $\det H = 0 \implies$ degenerate.

If a local minimum is not a global minimum, then either

- there is another local minimum which is the global minimum;
- the global minimum occurs on the boundary of f 's domain;
- or there is just no global minimum.

Example

Consider $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ given by $f(x, y) = x^3 + y^3 - 3xy$. We have

$$\begin{aligned} \nabla f &= (3x^2 - 3y, 3y^2 - 3x) \\ H &= \begin{pmatrix} 6x & -3 \\ -3 & 6y \end{pmatrix} \end{aligned}$$

Solving $\nabla f = \mathbf{0}$, we obtain $x^2 = y, y^2 = x \implies x^4 = x$, which gives $(x, y) = (0, 0)$ and $(x, y) = (1, 1)$.

Now, let us classify these points. We have $\det H = 9(4xy - 1), \text{Tr } H = 6(x + y)$.

- at $(x, y) = (1, 1)$, we have $\det H = 27 > 0, \text{Tr } H = 12 > 0$, so we have a local minimum, with $f(x, y) = -1$.
- at $(x, y) = (0, 0)$, we have $\det H = -9 < 0$, so we have a saddle point, with $f(x, y) = 0$. Furthermore, we have

$$\begin{aligned} \lambda_1 &= -3 & \mathbf{e}_1 &= \begin{pmatrix} 1 \\ 1 \end{pmatrix} \\ \lambda_2 &= 3 & \mathbf{e}_2 &= \begin{pmatrix} 1 \\ -1 \end{pmatrix} \end{aligned}$$

which indicates the directions in which f is increasing and decreasing.

However, there is no global maximum or minimum, since f increases/decreases without bound as $x^2 + y^2 \rightarrow \infty$.

§2 Convex functions

Definition 2.1 (Convex sets). A set $S \subseteq \mathbb{R}^n$ is called *convex* if, for every $\mathbf{x}, \mathbf{y} \in S$ and $\lambda \in [0, 1]$, the point $\lambda\mathbf{x} + (1 - \lambda)\mathbf{y} \in S$ – that is, every line segment with endpoints in S is contained in S .

Definition 2.2 (Graphs, chords and convexity). Let $f : D(f) \rightarrow \mathbb{R}$, where $D(f)$, its domain, is a subset of $\mathbb{R}^n$. The *graph* of f is the surface $z = f(\mathbf{x})$ in $\mathbb{R}^{n+1}$, whose points have coordinates $(\mathbf{x}, z)$. A *chord* of f is a line segment joining 2 points of its graph. Then f is *convex* if and only if

- $D(f)$ is convex;
- the graph of f lies below, or on, its chords – i.e., for any $\mathbf{x}, \mathbf{y} \in D(f)$ and $t \in [0, 1]$,

$$f((1 - t)\mathbf{x} + t\mathbf{y}) \leq (1 - t)f(\mathbf{x}) + tf(\mathbf{y})$$

where the LHS is well-defined since $D(f)$ is convex.

Furthermore, f is *strictly convex* if the above inequality is strict for $\mathbf{x} \neq \mathbf{y}$.

Definition 2.3 (Concavity). We say that f is (strictly) *concave* if and only if $-f$ is (strictly) convex.

Example

Let us have some examples.

- $f(x) = x^2$. Let us formally prove that this is (strictly) convex. We have

$$[(1 - t)x + ty]^2 - (1 - t)x^2 - ty^2 = -t(1 - t)(x - y)^2$$

which is strictly negative, as required.

- $f(x) = e^x$. This is also strictly convex.
- $f(x) = |x|$. This is convex, but not strictly.

§2.1 First order conditions

Theorem 2.4

If f is differentiable, then it is convex if and only if

$$f(\mathbf{y}) \geq f(\mathbf{x}) + (\mathbf{y} - \mathbf{x}) \cdot \nabla f(\mathbf{x})$$

for all $\mathbf{x}, \mathbf{y} \in D(f)$ – in other words, f lies above its tangent planes.

Proof. $\implies$ Let

$$\begin{aligned} h(t) &= (1 - t)f(\mathbf{x}) + tf(\mathbf{y}) - f((1 - t)\mathbf{x} + t\mathbf{y}) \\ \implies h'(0) &= -f(\mathbf{x}) + f(\mathbf{y}) - (\mathbf{y} - \mathbf{x}) \cdot \dot{\nabla} f(\mathbf{x}) \end{aligned}$$

However,

$$h'(0) = \lim_{h \rightarrow 0} \frac{h(t) - h(0)}{t} = \lim_{h \rightarrow 0} \frac{h(t)}{t}$$

By assumption, $h(t) \geq 0$ for all t , so $h'(0) \geq 0$, which gives the result.

$\Leftarrow$ We know

$$\begin{aligned} f(\mathbf{x}) &\geq f(\mathbf{z}) + (\mathbf{x} - \mathbf{z}) \cdot \nabla f(\mathbf{z}) \\ f(\mathbf{y}) &\geq f(\mathbf{z}) + (\mathbf{y} - \mathbf{z}) \cdot \nabla f(\mathbf{z}) \\ \implies (1-t)f(\mathbf{x}) + tf(\mathbf{y}) &\geq f(\mathbf{z}) + ((1-t)\mathbf{x} + t\mathbf{y} - \mathbf{z}) \cdot \nabla f(\mathbf{z}) \end{aligned}$$

Now, choose $\mathbf{z} = (1-t)\mathbf{x} + t\mathbf{y}$, which gives the result. ■

Corollary 2.5

If f is convex and differentiable, then any stationary point is a global minimum.

Proof. If $\mathbf{a}$ is a stationary point, then $\nabla f(\mathbf{a}) = \mathbf{0}$, so we immediately obtain $f(\mathbf{y}) \geq f(\mathbf{a})$ for all $\mathbf{y} \in D(f)$. ■

Proposition 2.6

The criterion in [theorem 2.4](#) is equivalent to

$$(\mathbf{y} - \mathbf{x}) \cdot (\nabla f(\mathbf{y}) - \nabla f(\mathbf{x})) \geq 0$$

for all $\mathbf{x}, \mathbf{y} \in D(f)$.

Remark. If $n = 1$, this tells us that $f'(y) \geq f'(x)$ if and only if $y \geq x$, in other words f' is monotonically increasing.

Proof. $\implies$ We have

$$\begin{aligned} f(\mathbf{y}) &\geq f(\mathbf{x}) + (\mathbf{y} - \mathbf{x}) \cdot \nabla f(\mathbf{x}) \\ f(\mathbf{x}) &\geq f(\mathbf{y}) + (\mathbf{x} - \mathbf{y}) \cdot \nabla f(\mathbf{y}) \end{aligned}$$

and adding these gives the result.

$\Leftarrow$ Let $\mathbf{z} = (1-t)\mathbf{x} + t\mathbf{y}$. Then

$$\begin{aligned} f(\mathbf{y}) - f(\mathbf{x}) &= [f(\mathbf{z})]_{t=0}^{t=1} \\ &= \int_0^1 \frac{d}{dt} f(\mathbf{z}) dt \\ &= \int_0^1 (\mathbf{y} - \mathbf{x}) \cdot \nabla f(\mathbf{z}) dt \end{aligned}$$

On the other hand,

$$f(\mathbf{y}) - f(\mathbf{x}) - (\mathbf{y} - \mathbf{x}) \cdot \nabla f(\mathbf{x}) = \int_0^1 (\mathbf{y} - \mathbf{x}) \cdot (\nabla f(\mathbf{z}) - \nabla f(\mathbf{x})) dt$$

Replacing $\mathbf{y}$ by $\mathbf{z}$ in the assumption, we obtain

$$t((\mathbf{y} - \mathbf{x}) \cdot (\nabla f(\mathbf{z}) - \nabla f(\mathbf{x}))) \geq 0$$

so the LHS is ≥ 0 and we obtain the result. ■

§2.2 Second order condition

Theorem 2.7

If $f \in C^2$, then it is convex if and only if the Hessian $H(\mathbf{x})$ is positive semi-definite, or equivalently has all eigenvalues ≥ 0 .

Proof. $\implies$ Use [proposition 2.6](#) writing $\mathbf{y} = \mathbf{x} + \mathbf{h}$ to obtain

$$\mathbf{h} \cdot (\nabla f(\mathbf{x} + \mathbf{h}) - \nabla f(\mathbf{x})) \geq 0$$

Taylor expanding, we can see

$$\nabla_i f(\mathbf{x} + \mathbf{h}) = \nabla_i f(\mathbf{x}) + h_j H_{ij}(\mathbf{x}) + O(h^2)$$

so, substituting this in, we obtain

$$h_i h_j H_{ij}(\mathbf{x}) + O(h^3) \geq 0$$

If $H(\mathbf{x})$ had some negative eigenvalue λ with eigenvector $\mathbf{e}$, we could make $\mathbf{h} = h\mathbf{e}$, giving

$$\lambda h^2 \mathbf{e}^2 + O(h^3) \geq 0$$

This is a clear contradiction for sufficiently small h .

$\Leftarrow$ For simplicity, let us take $n = 1$. Then

$$\begin{aligned} 0 &\leq \text{sign}(y - x) \int_x^y f''(z) dz = \text{sign}(y - x) (f'(y) - f'(x)) \\ \implies 0 &\leq (y - x) (f'(y) - f'(x)) \end{aligned}$$

so we conclude by [proposition 2.6](#). ■

Example

Let

$$f : \mathbb{R}^+ \times \mathbb{R}^+ \rightarrow \mathbb{R}$$

$$(x, y) \mapsto \frac{1}{xy}$$

with

$$H = \frac{1}{xy} \begin{pmatrix} \frac{2}{x^2} & \frac{1}{xy} \\ \frac{1}{xy} & \frac{2}{y^2} \end{pmatrix}$$

$$\det H = \frac{3}{x^4 y^4}$$

$$\text{Tr } H = \frac{2}{xy} \left(\frac{1}{x^2} + \frac{1}{y^2} \right)$$

so the eigenvalues of H are strictly positive, and hence f is strictly convex.

Importantly, if we had taken $D(f) = \{(x, y) : xy > 0\}$, the domain would not have been a convex set, so f would not have been convex.

§3 Legendre transform

Definition (Legendre transform). The *Legendre transform* of $f : D(f) \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ is given by

$$f^*(\mathbf{p}) = \sup_{\mathbf{x}} (\mathbf{p} \cdot \mathbf{x} - f(\mathbf{x}))$$

with domain the set of $\mathbf{p}$ such that this is finite.

Lemma 3.1

For any f , f^* is convex.

Proof. Let $\mathbf{p}, \mathbf{q} \in D(f^*)$, and $t \in (0, 1)$. Observe

$$\begin{aligned} & \sup_{\mathbf{x}} ((1-t)\mathbf{x} + t\mathbf{y}) \cdot \mathbf{x} - f(\mathbf{x}) \\ &= \sup_{\mathbf{x}} ((1-t)(\mathbf{p} \cdot \mathbf{x} - f(\mathbf{x})) + t(\mathbf{q} \cdot \mathbf{x} - f(\mathbf{x}))) \\ &\leq (1-t) \sup_{\mathbf{x}} (\mathbf{p} \cdot \mathbf{x} - f(\mathbf{x})) + t \sup_{\mathbf{x}} (\mathbf{q} \cdot \mathbf{x} - f(\mathbf{x})) \end{aligned}$$

Since the RHS is finite, the LHS is finite, so $(1-t)\mathbf{p} + t\mathbf{q} \in D(f^*)$ as required. Furthermore, we have also shown that

$$f^*((1-t)\mathbf{p} + t\mathbf{q}) \leq (1-t)f^*(\mathbf{p}) + tf^*(\mathbf{q})$$

which completes the proof. ■

Lemma 3.2

If f is convex, then so is

$$F_{\mathbf{p}}(\mathbf{x}) := f(\mathbf{x}) - \mathbf{p} \cdot \mathbf{x}$$

Proof. Clearly $D(F_{\mathbf{p}}) = D(f)$. Also,

$$\begin{aligned} F_{\mathbf{p}}((1-t)\mathbf{x} + t\mathbf{y}) &= f((1-t)\mathbf{x} + t\mathbf{y}) - \mathbf{p} \cdot ((1-t)\mathbf{x} + t\mathbf{y}) \\ &\leq (1-t)f(\mathbf{x}) + tf(\mathbf{y}) - \mathbf{p} \cdot (\cdots) \\ &= (1-t)F_{\mathbf{p}}(\mathbf{x}) + tF_{\mathbf{p}}(\mathbf{y}) \end{aligned}$$

■

Corollary 3.3

If f is convex and differentiable, then any stationary point of $\mathbf{p} \cdot \mathbf{x} - f(\mathbf{x})$ is a global maximum, occurring at $\mathbf{x}(\mathbf{p})$ obtained by solving

$$\nabla f(\mathbf{x}) = \mathbf{p}$$

Then, the Legendre transformation is $f^*(\mathbf{p}) = \mathbf{p} \cdot \mathbf{x}(\mathbf{p}) - f(\mathbf{x}(\mathbf{p}))$.

Remark. One can check that, if f is strictly convex, then the solution $\mathbf{x}(\mathbf{p})$ is unique.

Example

Let us check some examples.

- $f(x) = \frac{1}{2}ax^2$. For $a > 0$, this is strictly convex, and $x(p) = \frac{p}{a}$. Then

$$f^*(p) = \frac{p^2}{a} - \frac{p^2}{2a} = \frac{p^2}{2a}$$

- $f(v) = -\sqrt{1-v^2}$, $D(f) = (-1, 1)$. Then, solving for v , we have

$$p = \frac{v}{\sqrt{1-v^2}} \implies v = \frac{p}{\sqrt{1+p^2}}$$

so

$$f^*(p) = \frac{p^2}{\sqrt{1+p^2}} + \frac{1}{\sqrt{1+p^2}} = \sqrt{1+p^2}$$

- $f(x) = cx$, which is convex, but not strictly. Then

$$px - f(x) = (p - c)x$$

which has a supremum only if $p = c$. Hence $D(f^*) = \{c\}$, and $f^*(c) = 0$.

Theorem 3.4

If f is convex and C^2 , then $f^{**} = f$.

Proof. To determine f^{**} , we need to find $\mathbf{p}(\mathbf{x})$ satisfying $\nabla f^*(\mathbf{p}(\mathbf{x})) = \mathbf{x}$. Observe

$$\begin{aligned} f^*(\mathbf{p}) &= \mathbf{p} \cdot \mathbf{x}(\mathbf{p}) - f(\mathbf{x}(\mathbf{p})) \\ \Rightarrow \frac{\partial f^*}{\partial p_i} &= x_i + p_j \frac{\partial x_j}{\partial p_i} - \frac{\partial f}{\partial x_j} \bigg|_{\mathbf{x}=\mathbf{x}(\mathbf{p})} \frac{\partial x_j}{\partial p_i} \\ &= x_i + \underbrace{\left(p_j - \frac{\partial f}{\partial x_j} \bigg|_{\mathbf{x}=\mathbf{x}(\mathbf{p})} \right)}_0 \frac{\partial x_j}{\partial p_i} \\ \Rightarrow \nabla f^*(\mathbf{p}) &= \mathbf{x}(\mathbf{p}) \end{aligned}$$

We then obtain $\mathbf{x}(\mathbf{p}(\mathbf{x})) = \mathbf{x}$, i.e. $\mathbf{p}(\mathbf{x})$ is the inverse of $\mathbf{x}(\mathbf{p})$. Then

$$\begin{aligned} f^{**}(\mathbf{x}) &= \mathbf{x} \cdot \mathbf{p}(\mathbf{x}) - f^*(\mathbf{p}(\mathbf{x})) \\ &= \mathbf{x} \cdot \mathbf{p}(\mathbf{x}) - \left(\mathbf{p}(\mathbf{x}) \cdot \mathbf{x} - \underbrace{f(\mathbf{x}(\mathbf{p}(\mathbf{x})))}_{\mathbf{x}} \right) \\ &= f(\mathbf{x}) \end{aligned}$$

as required. ■

Remark. The convexity of f is clearly necessary, since f^{**} must be convex, being the Legendre transform of f^* . The C^2 condition, on the other hand, is not really necessary.

Example

Let us return to the example of $f(x) = cx$. Then

$$f^{**}(x) = \sup_{p \in \{c\}} (xp - f^*(p)) = cx = f(x)$$

as expected.

§3.1 Legendre transforms in thermodynamics

Consider a system with a few molecules, say 10^{23} . Suppose it is at thermal equilibrium, and define its macroscopic properties as follows:

- total energy E ;
- volume V ;
- temperature T ;
- pressure P .

Suppose also that the system is *isolated*, i.e. is not interacting with anything else. At equilibrium, the macroscopic physics of such a system is fully determined by E and V . And yet, there must be a huge number of microstates, $\Omega(E, V)$, all of which ‘look identical’ macroscopically. Indeed, $\Omega(E, V)$ is on the order of $10^{10^{23}}$! We quantify this with the following definition.

Definition 3.5 (Entropy). The *entropy*

$$S(E, V) := k_B \log \Omega(E, V)$$

Clearly, if E increases at fixed V , then there are more ways to partition E between molecules, so S should increase too. Hence, we should be able to invert to define $E(S, V)$. We now define

Definition 3.6 (Temperature and pressure). The *temperature* and *pressure* are defined as follows:

$$T := \left. \frac{\partial E}{\partial S} \right|_V$$

$$P := - \left. \frac{\partial E}{\partial V} \right|_S$$

Proposition 3.7 (Fundamental thermodynamic relation)

We have

$$dE = T dS - P dV$$

Proof. Apply the chain rule. ■

Now, forget [definition 3.6](#). Consider the Legendre transform of $E(S, V)$ with respect to S (at fixed V) with parameter T . Call this $-F$ – we have

$$-F(T, V) = \sup_S (TS - E(S, V))$$

As we will show on the example sheet, E is convex w.r.t. S , so we can evaluate the sup using [corollary 3.3](#). This gives that T should be the temperature as defined earlier! Indeed, we obtain

$$F(T, V) = E - TS$$

which is the *Helmholtz free energy*. Then we obtain

Proposition 3.8 (Properties of Helmholtz free energy)

We have

$$dF = -S dT - P dV$$

$$S = - \left. \frac{\partial F}{\partial T} \right|_V$$

$$P = - \left. \frac{\partial F}{\partial V} \right|_T$$

Remark. F is useful when considering a non-isolated system in equilibrium with an environment at temperature T .

Similarly, let us take the Legendre transform of $E(S, V)$ w.r.t. V (at fixed S) with parameter $-P$. Call this $-H(S, P)$:

$$-H(S, P) = \sup_V (-PV - E(S, V))$$

and the same argument for the convexity of E gives that P should be the pressure as defined earlier, and we obtain

$$H(S, P) = E + PV$$

which is the *enthalpy*.

Proposition 3.9 (Properties of enthalpy)

We have

$$\begin{aligned} dH &= T dS + V dP \\ T &= \left. \frac{\partial H}{\partial S} \right|_P \\ V &= \left. \frac{\partial H}{\partial P} \right|_S \end{aligned}$$

Remark. Enthalpy is useful when temperature and pressure are fixed, for example in chemistry.

Finally, let's take the Legendre transform of $E(S, V)$ with respect to both parameters at once, with parameters T and $-P$. Call this $-G(T, P)$.

Remark. This is the same as doing the Legendre transform twice, once with respect to S and once with respect to V .

We have

$$-G(T, P) = \sup_{S, V} (TS - PV - E(S, V))$$

Now, this is once again maximised when T is temperature and P is pressure, and we obtain

$$G(T, P) = E - TS + PV$$

which is the *Gibbs free energy*.

Proposition 3.10 (Properties of Gibbs free energy)

We have

$$\begin{aligned} dG &= -S dT + V dP \\ S &= - \left. \frac{\partial G}{\partial T} \right|_P \\ V &= \left. \frac{\partial G}{\partial P} \right|_T \end{aligned}$$

We have now recovered all four *thermodynamic potentials*.

Lecture 4 – 08/05/26

Let us consider some second derivatives. Observe

$$\begin{aligned} \left(\frac{\partial}{\partial S} \left(\frac{\partial E}{\partial V} \right)_S \right)_V &= \left(\frac{\partial}{\partial V} \left(\frac{\partial E}{\partial S} \right)_V \right)_S \\ \implies - \left(\frac{\partial P}{\partial S} \right)_V &= \left(\frac{\partial T}{\partial V} \right)_S \end{aligned}$$

and there are three similar equations that follow from doing the same thing with the other potentials, F , H and G . These are called the *Maxwell relations*.

§4 Constraints and Lagrange multipliers

Consider a smooth surface S in $\mathbb{R}^3$. If $f : \mathbb{R}^3 \rightarrow \mathbb{R}$, how do we determine the max/min value of f on S ?

Let (u, v) parameterise S , i.e. $\mathbf{x} = \mathbf{x}(u, v)$. We then need to minimise $f(\mathbf{x}(u, v))$, which we can just do by differentiating with respect to u and v . Indeed, we have

$$\begin{aligned} 0 &= \frac{\partial}{\partial u} f(\mathbf{x}(u, v)) = \frac{\partial \mathbf{x}}{\partial u} \cdot \nabla f \\ 0 &= \frac{\partial}{\partial v} f(\mathbf{x}(u, v)) = \frac{\partial \mathbf{x}}{\partial v} \cdot \nabla f \end{aligned}$$

Recall that $\{\frac{\partial \mathbf{x}}{\partial u}, \frac{\partial \mathbf{x}}{\partial v}\}$ are linearly independent vectors tangent to S , and so ∇f is normal to S . Hence, $\mathbf{x}_0 \in S$ is a stationary point of f on S if and only if $\nabla f(\mathbf{x}_0)$ is normal to S .

Often, however, we suffer from not having an explicit parameterisation $\mathbf{x}(u, v)$, and instead being limited to an implicit equation of the form $g(\mathbf{x}) = 0$.

Recall that ∇g is normal to such a surface S , and we already know that ∇f is normal to it, so we have

$$\nabla f(\mathbf{x}_0) = \lambda g(\mathbf{x}_0)$$

for some λ . Hence, define

$$\varphi(\mathbf{x}, \lambda) = f(\mathbf{x}) - \lambda g(\mathbf{x})$$

Observe

$$0 = \frac{\partial \varphi}{\partial \lambda} = 0$$

$$0 = \frac{\partial \varphi}{\partial \mathbf{x}} = \nabla f - \lambda \nabla g$$

so extremising φ is equivalent to solving our problem! Of course, these ideas could be generalised to $\mathbb{R}^n$.

Example (Direct method)

Suppose we would like to determine (x, y, z) that minimises surface area A for some open box of fixed volume $V = \frac{L^3}{2}$. In particular, we are minimising

$$A = 2z(x + y) + xy$$

subject to

$$xyz = \frac{L^3}{2}$$

Solving this directly, we have $z = \frac{L^3}{2xy}$, so

$$A = \frac{L^3}{x} + \frac{L^3}{y} + xy$$

which is stationary if

$$\begin{aligned} 0 = \frac{\partial A}{\partial x} &= -\frac{L^3}{x^2} + y \implies x^2 y = L^3 \\ 0 = \frac{\partial A}{\partial y} &= -\frac{L^3}{y^2} + x \implies xy^2 = L^3 \end{aligned}$$

This gives $x = y = L$, $z = \frac{L}{2}$, so $A = 3L^2$. We can compute the Hessian to check that this is the minimum.

Example (Lagrange multipliers)

Let us consider the same problem, and define

$$\varphi(x, y, z, \lambda) = A(x, y, z) - \lambda \left(xyz - \frac{L^3}{2} \right)$$

Hence

$$\begin{aligned} 0 = \frac{\partial \varphi}{\partial z} &= 2(x + y) - \lambda xy \implies \lambda = \frac{2(x + y)}{xy} \\ 0 = \frac{\partial \varphi}{\partial x} &= 2z + y - \lambda yz = 2z + y - \frac{2(x + y)z}{x} \\ &= \frac{y}{x}(x - 2z) \implies x = 2z \end{aligned}$$

and by symmetry, $y = 2z$ also. Finally,

$$0 = \frac{\partial \varphi}{\partial \lambda} = xyz - \frac{L^3}{2} = 4z^3 - \frac{L^3}{2}$$

which just recovers the constraint. Hence $z = \frac{L}{2}$, and $x = y = L$ as required.

However, to find the nature of the stationary point this way is not so easy, and in particular it does not correspond to the nature of the stationary point of φ . Observe

$$\frac{\partial^2 \varphi}{\partial \lambda^2} = 0$$

from which it's clear that H cannot be definite, so we always have a saddle point.

However, the clear advantage of this method over the direct one is that it does not require us to solve the constraint at any point.

Example

Let us minimise the quadratic form $f(\mathbf{x}) = x_i A_{ij} x_j$ on the surface $|\mathbf{x}|^2 = 1$, and let's not solve it directly! Have

$$\begin{aligned} \varphi(\mathbf{x}, \lambda) &= x_i A_{ij} x_j - \lambda(|\mathbf{x}|^2 - 1) \\ \implies 0 &= \frac{\partial \varphi}{\partial x_i} = A_{ij} x_j - \lambda x_i \end{aligned}$$

Hence, the stationary points are the normalised eigenvectors of A , and the Lagrange multiplier is the eigenvalue. Furthermore, evaluating f at a stationary point, we have

$$f = x_i A_{ij} x_j = \lambda x_i x_i = \lambda$$

and so the minimum of f is just given by the lowest eigenvalue.

Alternatively, we could have used a rotation to diagonalise A . Let us try out one more method.

Example

Define

$$\Lambda(\mathbf{x}) = \frac{f(\mathbf{x})}{g(\mathbf{x})} = \hat{x}_i A_{ij} \hat{x}_j = f(\hat{\mathbf{x}})$$

where $g(\mathbf{x}) = |\mathbf{x}|^2$. Then, minimising f with respect to $\hat{\mathbf{x}}$ is equivalent to minimising Λ with respect to $\mathbf{x}$. Observe:

$$\begin{aligned} 0 = \frac{\partial \Lambda}{\partial x_i} &= \frac{1}{g} \left(\nabla_i f - \frac{f}{g} \nabla_i g \right) = \frac{2}{g} \left(A_{ij} x_j - \frac{f}{g} x_i \right) \\ &= \frac{2}{g} (A_{ij} x_j - \Lambda x_i) \end{aligned}$$

so the extrema of Λ are the eigenvalues as required.

Example

Which probability distribution $\{p_1, \dots, p_n\}$ with $\sum p_i = 1$ maximises the information entropy

$$S = - \sum_i p_i \log_2 p_i$$

Define

$$\begin{aligned} \varphi(p_1, \dots, p_n, \lambda) &= - \sum_i p_i \log_2 p_i - \lambda \left(\sum p_i - 1 \right) \\ \implies 0 = \frac{\partial \varphi}{\partial p_i} &= - \log_2 p_i - \frac{1}{\ln 2} - \lambda \end{aligned}$$

from which it's immediately clear that all the p_i are equal, and hence each is $\frac{1}{n}$, giving $S_{max} = \log_2 n$.

Let us briefly discuss how to generalise these notions to multiple constraints.

Suppose we have $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and $m < n$ constraints $g_k(\mathbf{x}) = 0$ for $1 \leq k \leq m$. We introduce m Lagrange multipliers $\lambda_1, \dots, \lambda_m$ and extremise

$$\varphi(x_1, \dots, x_n, \lambda_1, \dots, \lambda_m) = f(\mathbf{x}) - \sum_{k=1}^m \lambda_k g_k(\mathbf{x})$$

§5 Functionals and the Euler-Lagrange equations

Definition 5.1 (Continuous differentiability and smoothness). For $\Omega \subset \mathbb{R}^n$, define

$$\begin{aligned} C^k(\Omega) &= \{y : \Omega \rightarrow \mathbb{R} : y \text{ is } k \text{ times continuously differentiable}\} \\ C^\infty(\Omega) &= \{y : \Omega \rightarrow \mathbb{R} : y \text{ is smooth}\} \end{aligned}$$

Definition 5.2 (Functionals). Let $\mathcal{C}$ be a set of functions, normally in $C^\infty(\Omega)$. A *functional* is a map

$$F : \mathcal{C} \rightarrow \mathbb{R}$$

$$y \mapsto F[y]$$

or indeed

$$\begin{aligned} F &: \mathcal{C}^n \rightarrow \mathbb{R} \\ (y_1, \dots, y_n) &\mapsto F[y_1, \dots, y_n] \end{aligned}$$

A particular important case that we will focus on is when $\Omega = [\alpha, \beta]$, $\mathcal{C} = C^\infty(\Omega)$, and

$$F[y] = \int_{\alpha}^{\beta} f(y(x), y'(x), x) dx$$

for some smooth f .

Which $y \in \mathcal{C}$ extremises f ? Consider varying y to $y + \delta y$ (where δy is of course a function). Then

$$\begin{aligned} \delta F[y] &= F[y + \delta y] - F[y] = \int_{\alpha}^{\beta} (f(y + \delta y, y' + \delta y', x) - f(y, y', x)) dx \\ &= \int_{\alpha}^{\beta} \left(\delta y \frac{\partial f}{\partial y} + \delta y' \frac{\partial f}{\partial y'} + \dots \right) dx \\ &= \left[\delta y \frac{\partial f}{\partial y'} \right]_{\alpha}^{\beta} + \int_{\alpha}^{\beta} \delta y(x) \left(\frac{\partial f}{\partial y} - \frac{d}{dx} \frac{\partial f}{\partial y'} \right) dx \end{aligned}$$

after integration by parts. We would really like to not bother with this boundary term, and so we will assume that functions in $\mathcal{C}$ have suitable boundary conditions for this term to vanish. For example, we could have

- $y(\alpha) = a, y(\beta) = b$ for any $y \in \mathcal{C}$, so $\delta y(\alpha) = \delta y(\beta) = 0$;
- $\frac{\partial f}{\partial y'} = 0$ at $x = \alpha, \beta$

Definition 5.3 (Functional derivative). We now define the *functional derivative* of $F[y]$ with respect to $y(x)$

$$\frac{\delta F[y]}{\delta y(x)} = \frac{\partial f}{\partial y} - \frac{d}{dx} \frac{\partial f}{\partial y'}$$

which is a function of x .

Now

$$\delta F[y] = \int_{\alpha}^{\beta} \delta y(x) \frac{\delta F[y]}{\delta y(x)} dx$$

In order to minimise F , then, recall that we want $\delta F[y]$ to vanish to first order for arbitrary $\delta y(x)$.

Theorem 5.4 (Euler-Lagrange equation)

It follows that F must satisfy the *Euler-Lagrange equation*

$$0 = \frac{\delta F[y]}{\delta y(x)} = \frac{\partial f}{\partial y} - \frac{d}{dx} \frac{\partial f}{\partial y'}$$

which is a second-order ODE for $y(x)$.

Let us also now note that we could define $\mathbf{y}(x) = (y_1(x), \dots, y_n(x))$,

$$F[\mathbf{y}] = \int_{\alpha}^{\beta} f(\mathbf{y}, \mathbf{y}', x) dx$$

We obtain

$$0 = \delta F[\mathbf{y}] = (\delta F[y_1], \dots, \delta F[y_n])$$

and so solving this system involves n simultaneous Euler-Lagrange equations.

§5.1 Geodesics of the Euclidean plane

What is the shortest path between two points A, B in Euclidean space? Recall

$$L = \int_C dl \qquad dl = \sqrt{dx^2 + dy^2}$$

Let us consider how to parameterise C . We could

- ① use x as a parameter and write $y = y(x)$, giving

$$L = \int_{x_a}^{x_b} \sqrt{1 + y'^2} dx$$

Observe $f = \sqrt{1 + y'^2}$ depends only on y' , so $\frac{\partial f}{\partial y} = 0$, meaning

$$0 = \frac{d}{dx} \frac{\partial f}{\partial y'} \implies \frac{\partial f}{\partial y'} = c$$

Now

$$\begin{aligned} \frac{y'}{\sqrt{1 + y'^2}} &= c \\ \implies y' &= m \end{aligned}$$

for some constant m . Hence $y = mx + c$, and m and c are fixed by boundary conditions, giving a line as required.

However, by using x as a parameter, we are excluding curves for which y is not a function of x . This motivates another parameterisation.

- ② use an arbitrary parameter t , with $\mathbf{x}(t) = (x(t), y(t))$. Then

$$L[\mathbf{x}] = \int_0^1 \sqrt{\dot{\mathbf{x}}(t)^2} dt$$

Now we have two Euler-Lagrange equations to solve. Looking first at the one for x , we have

$$\begin{aligned} 0 &= \underbrace{\frac{\partial f}{\partial x}}_0 - \frac{d}{dt} \frac{\partial f}{\partial \dot{x}} \implies \frac{\partial f}{\partial \dot{x}} = c \\ &\implies \frac{\dot{x}}{\sqrt{\dot{x}^2 + \dot{y}^2}} = c \end{aligned}$$

Similarly, the Euler-Lagrange equation for y gives

$$\frac{\dot{y}}{\sqrt{\dot{x}^2 + \dot{y}^2}} = s$$

so, squaring each and taking the sum, we have

$$c^2 + s^2 = 1 \implies c = \cos \theta, s = \sin \theta$$

for some θ .

Now, it will be desirable to fix our parameterisation to in fact be by arc length. We have

$$\cos \theta = \frac{\dot{x}}{\sqrt{\dot{x}^2 + \dot{y}^2}} = \frac{dx}{dt} \frac{dt}{dl} = \frac{dx}{dl}$$

and analogously for y . Hence

$$x = x_A + l \cos \theta \qquad y = y_A + l \sin \theta$$

as required.

Importantly, we couldn't have used l as our original parameter, since, for different curves, it has a different upper limit – it was necessary to apply Euler-Lagrange with t , and only then change to l .

§6 First integrals and Fermat's principle

Recall the Euler-Lagrange equation:

$$0 = \frac{\partial f}{\partial y} - \frac{d}{dx} \frac{\partial f}{\partial y'}$$

If $\frac{\partial f}{\partial y} = 0$, then we can integrate directly to get $\frac{\partial f}{\partial y'} = \text{const.}$

Lemma 6.1

Furthermore, if f has no explicit x -dependence, in other words $\frac{\partial f}{\partial x} = 0$, then we claim

$$f - y' \frac{\partial f}{\partial y'} = \text{const.}$$

for a solution y (note f still depends implicitly on x via y and y')

Proof. Observe

$$\begin{aligned} \frac{df}{dx} &= \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} y' + \frac{\partial f}{\partial y'} y'' \\ &= \frac{\partial f}{\partial x} + y' \left(\frac{\partial f}{\partial y} - \frac{d}{dx} \frac{\partial f}{\partial y'} \right) + \frac{d}{dx} \left(y' \frac{\partial f}{\partial y'} \right) \end{aligned}$$

If the Euler-Lagrange equation holds, the contents of the second bracket is zero, giving

$$\frac{d}{dx} \left(f - y' \frac{\partial f}{\partial y'} \right) = \frac{\partial f}{\partial x}$$

giving the result. ■

§6.1 Fermat's principle

Consider light propagating in a medium, and let the speed in that medium be given by $c(\mathbf{x})$. Define the *refractive index*

$$n(\mathbf{x}) = \frac{c}{c(\mathbf{x})}$$

where c is the speed of light in a vacuum. Then, the time for light to travel between points A and B along a path C is

$$T[C] = \int_C \frac{1}{c(\mathbf{x})} dl = \frac{1}{c} \underbrace{\int_C n(\mathbf{x}) dl}_{P[C]}$$

where $P[C]$ is called the *optical path length*.

Proposition 6.2 (Fermat's principle)

Light travels from A to B along a path that locally minimises T or P .

Example

Let us find the path of a light ray in the (x, z) plane in a medium with

$$n(z) = \sqrt{a - bz}$$

for constants a, b . For simplicity, assume a path C may be parameterised as $z(x)$ by $\alpha \leq x \leq \beta$. Then

$$P[z] = \int_{\alpha}^{\beta} n(z) \sqrt{1 + z'^2} dx = \int_{\alpha}^{\beta} f(z, z') dx$$

Observe we lack any explicit x -dependence, so we know

$$\begin{aligned} k &= f - z' \frac{\partial f}{\partial z'} = \frac{n(z)}{\sqrt{1 + z'^2}} = \sqrt{\frac{a - bz}{1 + z'^2}} \\ \implies z'^2 &= \frac{b}{k^2} (z_0 - z) & z_0 &= \frac{a - k^2}{b} \\ \implies z' &= \pm \sqrt{\frac{b}{k^2} (z_0 - z)} \end{aligned}$$

This is a separable differential equation – we obtain

$$z = z_0 - \frac{b}{4k^2} (x - x_0)^2$$

which is a parabola.

Example (The brachistochrone!)

A bead slides on a frictionless wire in a vertical plane. What shape of wire minimises the time for the bead to fall from rest at A to a lower, horizontally displaced, point B ?

Since the wire is frictionless, we may apply energy conservation and, making A the origin, we have

$$\frac{1}{2}mv^2 + mgy = 0$$

so $v = \sqrt{-2gy}$. Then

$$T = \int_A^B \frac{1}{v} dl = \frac{1}{\sqrt{2g}} \int_A^B \frac{dx^2 + dy^2}{\sqrt{-y}}$$

Assume for simplicity that the path may be parameterised by x , then

$$T \propto \int_0^{x_B} \frac{\sqrt{1 + y'^2}}{\sqrt{-y}} dx$$

Once again, f has no explicit x -dependence, so there exists a first integral given by

$$\begin{aligned} k &= \frac{1}{\sqrt{-y(1 + y'^2)}} \\ \implies 2c &= -y(1 + y'^2) \end{aligned} \quad (*)$$

for some constant $c \geq 0$. The simplest way to solve this equation is to substitute

$$y = -c(1 - \cos \theta) \quad \theta \geq 0$$

which will give us

$$\begin{aligned} \frac{dx}{d\theta} &= \pm c(1 - \cos \theta) \\ \implies x &= c(\theta - \sin \theta) \end{aligned}$$

where we have chosen c positive for x to be increasing, and applying that C passes through the origin. Requiring the curve to pass through $B = (x_B, y_B)$ fixes c and the value of θ at B .

§7 Constrained variation of functionals

How can we extremise a functional $F[y]$ subject to a functional constraint $P[y] = c$ constant? We can use a Lagrange multiplier! Define

$$\varphi_\lambda[y] = F[y] - \lambda(P[y] - c)$$

and we want to extremise this with respect to y (recall differentiating with respect to λ just recovers the constraint). Hence, we obtain

$$\frac{\delta F}{\delta y(x)} - \lambda \frac{\delta P}{\delta y(x)} = 0$$

Example

Let us try to minimise the functional

$$P[y] = \int_0^1 \frac{1}{2} y'^2 dx$$

$$\int_0^1 y(x) dx = 1$$

over smooth functions on $[0, 1]$ with $y(0) = y(1) = 0$. We obtain

$$\begin{aligned} \varphi_\lambda[y] &= \int_0^1 \frac{1}{2} y'^2 dx - \lambda \left(\int_0^1 y dx - 1 \right) \\ &= \int_0^1 \left(\frac{1}{2} y'^2 - \lambda y \right) dx + \lambda \end{aligned}$$

Extremising this, we obtain the Euler-Lagrange equation

$$\begin{aligned} 0 &= \frac{\partial f}{\partial y} - \frac{d}{dx} \frac{\partial f}{\partial y'} = -\lambda - y'' \\ \implies y &= -\frac{1}{2} \lambda x^2 + Ax + B \end{aligned}$$

where the values at the endpoints will give the values of A and B . Finally, substituting back into the constraint will give

$$1 = \int_0^1 y dx = \left[-\frac{1}{6} \lambda x^3 + \frac{1}{2} \left(\frac{1}{2} \lambda \right) x^2 \right]_0^1 = \frac{\lambda}{12}$$

so $\lambda = 12$ (and $B = 0$, $A = \frac{1}{2} \lambda$).

Example (Dido's problem)

Given a straight wall and a fence of length L , what shape of fence maximises the enclosed area? See example sheet 1.

Example (Isoperimetric problem)

Given a fence of length L , what shape of fence maximises the enclosed area? See example sheet 2.

§7.1 Sturm-Liouville problem

Consider

$$\mathcal{C} = \{y : [\alpha, \beta] \rightarrow \mathbb{R} : y \in C^2, y(\alpha) = y(\beta) = 0\}$$

and let

$$F[y] = \int_\alpha^\beta (\rho(x) y'(x)^2 + \sigma(x) y(x)^2) dx$$

$$G[y] = \int_{\alpha}^{\beta} w(x)y(x)^2 dx$$

where ρ , σ and w are known, ρ and w are positive on (α, β) , and we would like to minimise $F[y]$ subject to the constraint $G[y] = 1$.

Let us introduce a Lagrange multiplier – we need to extremise

$$\varphi_{\lambda}[y] = F[y] - \lambda(G[y] - 1)$$

so we obtain

$$0 = \frac{\delta F[y]}{\delta y(x)} - \lambda \frac{\delta G[y]}{\delta y(x)}$$

Observe

$$\begin{aligned} \frac{\delta F[y]}{\delta y(x)} &= 2\sigma y - \frac{d}{dx} (2\rho y') =: 2\mathcal{L}y \\ \frac{\delta G[y]}{\delta y(x)} &= 2wy \end{aligned}$$

so we obtain

$$\mathcal{L}y = \lambda wy$$

so we have obtained an eigenvalue problem. With the boundary conditions $y(\alpha) = y(\beta) = 0$, the solution will be trivial for all but a small collection of λ . Note that, if $y(x)$ solves the equation, so does $Cy(x)$ – the constraint $G[y] = 1$ is essentially normalising y . Now

$$\begin{aligned} F[y] &= - \underbrace{[\rho y y']_{\alpha}^{\beta}}_0 + \int_{\alpha}^{\beta} y \mathcal{L}y dx \\ &= \lambda \int_{\alpha}^{\beta} w y^2 dx \\ &= \lambda G[y] = \lambda \end{aligned}$$

Hence, the solution is given by the lowest eigenvalue of $\mathcal{L}$. Note, by the way, that we can change the boundary conditions to any combination of y and y' being zero at the endpoints, and this will still work.

Another approach would be to define

$$\begin{aligned} \Lambda[y] &= \frac{F[y]}{G[y]} \\ \implies 0 &= \frac{\delta \Lambda[y]}{\delta y(x)} = \frac{1}{G} \left(\frac{\delta F[y]}{\delta y(x)} - \frac{F}{G} \frac{\delta G[y]}{\delta y(x)} \right) \\ &= \frac{2}{G} (\mathcal{L}y - \Lambda wy) \end{aligned}$$

recovering the same solution.

§7.2 Geodesics on a surface and function constraints

Consider a surface S in Euclidean space with equation $g(\mathbf{x}) = 0$. Let $A, B \in S$ – what is the shortest curve between A and B on S ? Let $\mathbf{x}(t)$, $0 \leq t \leq 1$ be a curve on S , with $\mathbf{x}(0) = \mathbf{x}_A$, $\mathbf{x}(1) = \mathbf{x}_B$.

We want to extremise the functional

$$L(\mathbf{x}) = \int_0^1 \sqrt{\dot{\mathbf{x}}^2} dt$$

subject to

$$g(\mathbf{x}(t)) = 0$$

In order to do this, we need to define a Lagrange multiplier *function* $\lambda(t)$, and let

$$\varphi[\mathbf{x}, \lambda] = L[\mathbf{x}] - \int_0^1 \lambda(t) g(\mathbf{x}(t)) dt$$

Extremising with respect to λ , we obtain

$$0 = \frac{\delta \varphi}{\delta \lambda(t)} = -g(\mathbf{x}(t)) \quad (*)$$

which recovers the constraint as we would hope. Extremising with respect to x_i , we have

$$0 = \frac{\delta \varphi}{\delta x_i(t)} = -\lambda \nabla_i g(\mathbf{x}(t)) - \frac{d}{dt} \left(\frac{\dot{x}_i}{\sqrt{\dot{\mathbf{x}}^2}} \right) \quad (\dagger)$$

How do we solve this? Taking the time derivative of $(*)$, we obtain

$$\begin{aligned} 0 = \dot{x}_i \partial_i g &\implies \frac{\dot{x}_i \partial_i g}{\sqrt{\dot{\mathbf{x}}^2}} = 0 \\ \implies 0 &= \frac{d}{dt} \left(\frac{\dot{x}_i}{\sqrt{\dot{\mathbf{x}}^2}} \right) \partial_i g + \frac{\dot{x}_i \dot{x}_j \partial_i \partial_j g}{\sqrt{\dot{\mathbf{x}}^2}} \end{aligned}$$

By $(\dagger)$, the first term is $-\lambda(\nabla g)^2$, so

$$\lambda(t) = \frac{\dot{x}_i \dot{x}_j \partial_i \partial_j g}{(\nabla g)^2 \sqrt{\dot{\mathbf{x}}^2}}$$

Finally, then, dividing $(\dagger)$ by $\sqrt{\dot{\mathbf{x}}^2}$, we obtain

$$\boxed{\frac{1}{\sqrt{\dot{\mathbf{x}}^2}} \frac{d}{dt} \left(\frac{\dot{x}_i}{\sqrt{\dot{\mathbf{x}}^2}} \right) + \frac{\dot{x}_j \dot{x}_k \partial_j \partial_k g}{(\nabla g)^2 \sqrt{\dot{\mathbf{x}}^2}} \nabla_i g = 0}$$

This is the equation for the geodesics on the surface, to be solved along with $(*)$. However, we can simplify this heavily by using arc length as a parameter – since $\frac{d}{dt} = \frac{1}{\sqrt{\dot{\mathbf{x}}^2}} \frac{d}{dl}$, we obtain

$$\frac{d^2 x_i}{dl^2} + \frac{1}{(\nabla g)^2} \frac{dx_j}{dl} \frac{dx_k}{dl} \partial_j \partial_k g \nabla_i g = 0$$

Example

As a sanity check, let us consider the surface $g = z$, i.e. the plane $z = 0$. Then $\partial_j \partial_k g = 0$, so $\frac{d^2 x_i}{dl^2} = 0$, and we obtain a straight line as expected.

§8 Principle of least action

Consider a system of N particles in 3D. The positions $\mathbf{x}_1, \dots, \mathbf{x}_N$ correspond to a single point $(\mathbf{x}_1, \dots, \mathbf{x}_N)$ in a $3N$ -dimensional *configuration space*. Recall that it is sometimes preferable to use non-Cartesian coordinates $\mathbf{q} = (\mathbf{q}_1, \dots, \mathbf{q}_N)$ for this space.

If we know the kinetic energy T and potential energy V in terms of $\mathbf{q}$ and $\dot{\mathbf{q}}$, then we can easily determine the equations of motion.

Definition 8.1 (Lagrangian). The *Lagrangian* for a system is given by

$$L(\mathbf{q}, \dot{\mathbf{q}}, t) = T - V$$

Definition 8.2 (Action). The *action* of a path $\mathbf{q}(t)$ in configuration space between $\mathbf{q}_A$ at time t_A and $\mathbf{q}_B$ at time t_B is given by

$$I[\mathbf{q}] = \int_{t_A}^{t_B} L(\mathbf{q}, \dot{\mathbf{q}}, t) dt$$

Remark. The dimensions of I are energy multiplied by time, i.e. ML^2T^{-1} , notably the same as those of Planck's constant from quantum mechanics. Indeed, the path integral formulation of quantum mechanics produces, in the classical limit, the principle of least action.

Proposition 8.3 (Principle of least action)

The physical path taken from $\mathbf{q}_A$ to $\mathbf{q}_B$ is the one which extremises the action.

Since $\mathbf{q}_A$ and $\mathbf{q}_B$ are fixed, the boundary term vanishes, so the path satisfies the Euler-Lagrange equations:

$$\frac{\partial L}{\partial q_i} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} = 0 \quad i = 1, \dots, 3N$$

Example

Take $N = 1$, $\mathbf{q} = \mathbf{x}$, and

$$\begin{aligned} T &= \frac{1}{2}m\dot{\mathbf{x}}^2 & V &= V(\mathbf{x}, t) \\ \implies L &= \frac{1}{2}m\dot{\mathbf{x}}^2 - V = V(\mathbf{x}, t) \end{aligned}$$

Hence Euler-Lagrange gives

$$-\frac{\partial V}{\partial x_i} - \frac{d}{dt}(m\dot{x}_i) = 0$$

or, in vector notation,

$$m\ddot{\mathbf{x}} = -\nabla V$$

which is Newton's second law for a conservative force $\mathbf{F} = -\nabla V$.

Proposition 8.4

If L has no explicit t -dependence, then the Euler-Lagrange equations admit a first integral

$$L - \sum_{i=1}^{3N} \dot{q}_i \frac{\partial L}{\partial \dot{q}_i} = -E$$

where $-E$ is a constant.

Proof. Have

$$\begin{aligned} \frac{dL}{dt} &= \frac{d}{dt}L(\mathbf{q}(t), \dot{\mathbf{q}}(t), t) \\ &= \frac{\partial L}{\partial t} + \sum_{i=1}^{3N} \left(\frac{\partial L}{\partial q_i} \dot{q}_i + \frac{\partial L}{\partial \dot{q}_i} \ddot{q}_i \right) \\ &= \frac{\partial L}{\partial t} + \sum_{i=1}^{3N} \dot{q}_i \underbrace{\left(\frac{\partial L}{\partial \dot{q}_i} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} \right)}_0 + \frac{d}{dt} \sum_{i=1}^{3N} \dot{q}_i \frac{L}{\dot{q}_i} \\ \implies 0 &= \frac{\partial L}{\partial t} = \frac{d}{dt} \left(L - \sum_{i=1}^{3N} \dot{q}_i \frac{\partial L}{\partial \dot{q}_i} \right) \end{aligned}$$

so this quantity is a constant as required. ■

Example

Suppose $V = V(\mathbf{x})$, so this result applies. We obtain

$$-E = \frac{1}{2}m\dot{x}^2 - V - \sum_i \dot{x}_i m \dot{x}_i = -(T + V)$$

so E is the total energy.

Remark. Why is energy conserved? It is a consequence of the lack of dependence of the Lagrangian on time, which itself represents the fact that the laws of physics do not change over time. This is a special case of Noether's theorem.

§8.1 Central force field

Consider a particle in 2D, and use polar coordinates. Assume $V = V(r)$, so the Lagrangian is given by

$$L = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\varphi}^2) - V(r)$$

The Euler-Lagrange equations give

$$\begin{aligned} 0 &= \underbrace{\frac{\partial L}{\partial \varphi}}_0 - \frac{d}{dt} \frac{\partial L}{\partial \dot{\varphi}} \\ \implies \frac{\partial L}{\partial \dot{\varphi}} &= mr^2\dot{\varphi} = mh \end{aligned}$$

where mh is a constant. Recall also that, since V is time-independent, we have a conserved energy

$$E = \dot{r} + \frac{\partial L}{\partial \dot{r}} + \dot{\varphi} \frac{\partial L}{\partial \dot{\varphi}} - L = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\varphi}^2) + V(r)$$

Now, eliminating $\dot{\varphi}$, we obtain

$$\frac{1}{2}m\dot{r}^2 + V_{\text{eff}}(r) = E$$

where $V_{\text{eff}}(r)$, the effective potential, is $\frac{mh^2}{2r^2} + V(r)$.

§8.2 The Hamiltonian and Hamilton's equations

Assume that $L(\mathbf{q}, \dot{\mathbf{q}}, t)$ is a convex function of $\dot{\mathbf{q}}$. Then, we can easily take the Legendre transform with respect to $\dot{\mathbf{q}}$ to obtain the *Hamiltonian*

$$\begin{aligned} H(\mathbf{q}, \mathbf{p}, t) &= \sup_{\dot{\mathbf{q}}} (\mathbf{p} \cdot \dot{\mathbf{q}} - L(\mathbf{q}, \dot{\mathbf{q}}, t)) \\ &= (\mathbf{p} \cdot \dot{\mathbf{q}} - L(\mathbf{q}, \dot{\mathbf{q}}, t))|_{\dot{\mathbf{q}}=\dot{\mathbf{q}}^*} \end{aligned}$$

where $\dot{\mathbf{q}}^*$ is the stationary point of $\mathbf{p} \cdot \dot{\mathbf{q}} - L(\mathbf{q}, \dot{\mathbf{q}}, t)$. Indeed, we obtain

Proposition 8.5

$$p_i = \frac{\partial L}{\partial \dot{q}_i}$$

Example

Take $N = 1$, $\mathbf{q} = \mathbf{x}$, $L = \frac{1}{2}m\dot{\mathbf{x}}^2 - V(\mathbf{x}, t)$. To find the Hamiltonian, solve for the stationary point of $\dot{\mathbf{x}}$,

$$p_i = \frac{\partial L}{\partial \dot{x}_i} = m\dot{x}_i$$

so that

$$H = \mathbf{p} \cdot \frac{\mathbf{p}}{m} - \left(\frac{\mathbf{p}^2}{2m} - V \right) = \frac{\mathbf{p}^2}{2m} + V$$

which is the system's total energy.

Example (Central force)

Take $N = 1$, and work in 2D polars, so $L = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\phi}^2) - V(r)$. Solving for the stationary point, we obtain

$$\begin{aligned} p_r = \frac{\partial L}{\partial \dot{r}} &= m\dot{r} & \implies \dot{r} &= \frac{p_r}{m} \\ p_\phi = \frac{\partial L}{\partial \dot{\phi}} &= mr^2\dot{\phi} & \implies \dot{\phi} &= \frac{p_\phi}{mr^2} \end{aligned}$$

We obtain

$$\begin{aligned} H &= p_r \frac{p_r}{m} + p_\phi \frac{p_\phi}{mr^2} - L \\ &= \frac{p_r^2}{2m} + \frac{p_\phi^2}{2mr^2} + V(r) \end{aligned}$$

which is again the total energy.

In general, p_i is called the *conjugate momentum* to q_i , and we can importantly construct this in any coordinate system.

Furthermore, we see that

$$H = \sum_i \dot{q}_i \frac{\partial L}{\partial \dot{q}_i} - L$$

so the Hamiltonian really is the total energy – *but*, with $\dot{\mathbf{q}}$ eliminated in favour of $\mathbf{q}$, $\mathbf{p}$ and t using $p_i = \frac{\partial L}{\partial \dot{q}_i}$. We see that, if $\frac{\partial L}{\partial t} = 0$, H is conserved.

Once we know about the Hamiltonian, we can actually forget about the Lagrangian and write the equations of motion solely in terms of H . Observe

$$\left(\frac{\partial H}{\partial p_i} \right)_{\mathbf{q}, t} = \dot{q}_i + \sum_j p_j \frac{\partial \dot{q}_j}{\partial p_i} - \sum_j \frac{\partial L}{\partial \dot{q}_j} \frac{\partial \dot{q}_j}{\partial p_i}$$

$$= \dot{q}_i$$

since the second and third terms cancel by [proposition 8.5](#). Also,

$$\begin{aligned} \left(\frac{\partial H}{\partial q_i} \right)_{\mathbf{p}, t} &= \sum_j p_j \frac{\partial \dot{q}_j}{\partial q_i} - \frac{\partial L}{\partial q_i} - \sum_j \frac{\partial L}{\partial \dot{q}_j} \frac{\partial \dot{q}_j}{\partial q_i} \\ &= -\frac{\partial L}{\partial q_i} = -\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} \quad (\text{Euler-Lagrange}) \\ &= -\dot{p}_i \end{aligned}$$

where again we have cancellation by [proposition 8.5](#). We obtain

Proposition 8.6 (Hamilton's equations)

We have

$$\dot{q}_i = \left(\frac{\partial H}{\partial p_i} \right)_{\mathbf{q}, t} \quad \dot{p}_i = - \left(\frac{\partial H}{\partial q_i} \right)_{\mathbf{p}, t}$$

Recall that the Euler-Lagrange equations are second order ODEs in a $3N$ -dimensional configuration space. On the other hand, Hamilton's equations are first order ODEs but in six-dimensional *phase space* with coordinates $(\mathbf{q}, \mathbf{p})$.

Let us derive these equations another way. Parameterise the action by both $\mathbf{p}$ and $\mathbf{q}$, to obtain

$$I[\mathbf{p}, \mathbf{q}] = \int_{t_A}^{t_B} (\mathbf{p} \cdot \dot{\mathbf{q}} - H(\mathbf{q}, \mathbf{p}, t))$$

using the Lagrangian-Hamiltonian relationship from earlier, and where $\mathbf{q}(t_A)$, $\mathbf{q}(t_B)$ are fixed. Then, minimising this new functional will recover Hamilton's equations.

Remark. Recall

$$\begin{aligned} \frac{dH}{dt} &= \frac{d}{dt} (H(\mathbf{q}, \mathbf{p}, t)) \\ &= \sum_i \left(\frac{\partial H}{\partial q_i} \dot{q}_i + \frac{\partial H}{\partial p_i} \dot{p}_i \right) + \frac{\partial H}{\partial t} \\ &= \frac{\partial H}{\partial t} \end{aligned}$$

by Hamilton's equations. Hence, if H has no *explicit* t -dependence, then it has no t -dependence at all – that is, it is conserved.

§9 Symmetries and Noether's theorem

Informally, Noether's theorem states that continuous symmetries of the action correspond to first integrals (i.e. conservation laws) of the Euler-Lagrange equations.

Definition 9.1 (Symmetries of the action). Define a smooth map

$$\mathbf{Q} : \mathbb{R}^{6n+1} \rightarrow \mathbb{R}^{3n}$$

which we should think of as going from Hamiltonian phase space to Lagrangian configuration space. Given a curve $\mathbf{q}(t)$, define

$$\mathbf{q}^*(t) = \mathbf{Q}(\mathbf{q}(t), \dot{\mathbf{q}}(t), t)$$

If, for any curve $\mathbf{q}(t)$ and any t_A, t_B ,

$$I[\mathbf{q}^*] = \int_{t_A}^{t_B} L(\mathbf{q}^*(t), \dot{\mathbf{q}}^*(t), t) dt = \int_{t_A}^{t_B} L(\mathbf{q}(t), \dot{\mathbf{q}}(t), t) dt = I[\mathbf{q}]$$

then $\mathbf{Q}$ is an *symmetry of the action*.

Definition 9.2 (Continuous symmetries of the action). Consider a parameterised family of symmetries of the action,

$$\mathbf{Q}(\mathbf{q}, \dot{\mathbf{q}}, t, s)$$

such that $\mathbf{Q}(\mathbf{q}, \dot{\mathbf{q}}, t, 0) = \mathbf{q}$, that is, for $s = 0$, we have the identity map. Such a family is called a continuous symmetry of the action.

Theorem 9.3 (Noether's theorem, 1)

If there exists a 1-parameter family of symmetries of the action, then any solution of the Euler-Lagrange equations has a conserved quantity given by

$$\sum_{i=1}^{3N} \frac{\partial L}{\partial \dot{q}_i} \left(\frac{\partial Q_i}{\partial s} \right)_{s=0}$$

Proof. Take s small, so we may Taylor expand $\mathbf{Q}$ to obtain

$$\mathbf{q}^* = \mathbf{q} + s \left(\frac{\partial \mathbf{Q}}{\partial s} \right)_{s=0} + O(s^2)$$

Then, substituting this into the definition of a symmetry of the action, we obtain

$$\begin{aligned} 0 &= \int_{t_A}^{t_B} \left(L(\mathbf{q} + s \left(\frac{\partial \mathbf{Q}}{\partial s} \right)_{s=0} + \dots, \dot{\mathbf{q}} + s \frac{d}{dt} \left(\frac{\partial \mathbf{Q}}{\partial s} \right)_{s=0} + \dots, t) - L(\mathbf{q}, \dot{\mathbf{q}}, t) \right) dt \\ &= \int_{t_A}^{t_B} \left(s \left(\frac{\partial Q_i}{\partial s} \right)_{s=0} \frac{\partial L}{\partial q_i} + s \frac{d}{dt} \left(\frac{\partial Q_i}{\partial s} \right)_{s=0} \frac{\partial L}{\partial \dot{q}_i} + \dots \right) dt \\ &= \int_{t_A}^{t_B} \left(s \left(\frac{\partial Q_i}{\partial s} \right)_{s=0} \underbrace{\left(\frac{\partial L}{\partial q_i} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} \right)}_{0 \text{ by E-L}} + s \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \left(\frac{\partial Q_i}{\partial s} \right)_{s=0} \right) + \dots \right) dt \\ &= \left[s \frac{\partial L}{\partial \dot{q}_i} \left(\frac{\partial Q_i}{\partial s} \right)_{s=0} \right]_{t_A}^{t_B} + O(s^2) \end{aligned}$$

Hence,

$$\frac{\partial L}{\partial \dot{q}_i} \left(\frac{\partial Q_i}{\partial s} \right)_{s=0}$$

takes the same value at every time, as required. ■

Example

Say L doesn't depend on $\mathbf{q}$, then consider

$$\mathbf{Q}(\mathbf{q}) = (q_1 + s, \dots, q_{3N})$$

so

$$\begin{aligned} q_1^* &= q_1 + s & \cdots & & q_{3N}^* &= q_{3N} \\ \dot{q}_1^* &= \dot{q}_1 & \cdots & & \dot{q}_{3N}^* &= \dot{q}_{3N} \end{aligned}$$

but, since L doesn't depend on $\mathbf{q}$, this is a symmetry of the action. Hence, by Noether's theorem, we have the conserved quantity

$$\frac{\partial L}{\partial \dot{q}_i} \left(\frac{\partial Q_i}{\partial s} \right)_{s=0} = \frac{\partial L}{\partial \dot{q}_i} = p_i$$

so the conjugate momentum is conserved.

Lecture $n + 1$ **Example**

Let us use Cartesian coordinates, $\mathbf{q} = (\mathbf{x}_1, \dots, \mathbf{x}_n)$. Suppose L depends only on the pairwise differences between the $\mathbf{x}_i$. Define

$$\mathbf{Q}(\mathbf{q}) = (\mathbf{x}_1 + s\mathbf{a}, \dots, \mathbf{x}_N + s\mathbf{a})$$

Clearly this preserves the pairwise differences, so this is a continuous symmetry of the action. Hence, by Noether's theorem, we have the conserved quantity

$$\sum_i \frac{\partial L}{\partial \dot{\mathbf{q}}_i} \left(\frac{\partial Q_i}{\partial s} \right)_{s=0} = \mathbf{a} \cdot \sum_i \frac{\partial L}{\partial \dot{\mathbf{x}}_i} = \mathbf{a} \cdot \sum_i \mathbf{p}_i = \mathbf{a} \cdot \mathbf{P}$$

where $\mathbf{P}$ is the total momentum. Thus, since $\mathbf{a}$ is arbitrary, $\mathbf{P}$ is conserved. In other words, invariance of physical laws under spatial translations implies conservation of momentum.

Example

Let us now consider a rotation R , and the corresponding transformation $\mathbf{Q}$ which maps

$$\mathbf{Q}(\mathbf{q}) = (R\mathbf{x}_1, \dots, R\mathbf{x}_n)$$

In particular, let us take $R = R(s)$, a rotation through angle s about some fixed axis $\hat{\mathbf{n}}$. For infinitesimal s , $R(s)\mathbf{x} = \mathbf{x} + s\hat{\mathbf{n}} \times \mathbf{x}$. Hence

$$\left(\frac{\partial Q_i}{\partial s} \right)_{s=0} = (\hat{\mathbf{n}} \times \mathbf{x}_1, \dots, \hat{\mathbf{n}} \times \mathbf{x}_N)$$

Thus, our conserved quantity is given by

$$\sum_{i,a} \frac{\partial L}{\partial (\dot{x}_i)_a} (\hat{\mathbf{n}} \times \mathbf{x}_i)_a = \sum_i \mathbf{p}_i \cdot (\hat{\mathbf{n}} \times \mathbf{x}_i) = \hat{\mathbf{n}} \cdot \sum_i \mathbf{x}_i \times \mathbf{p}_i = \hat{\mathbf{n}} \cdot \mathbf{L}$$

so, since $\hat{\mathbf{n}}$ is arbitrary, total angular momentum $\mathbf{L}$ is conserved.

We can generalise our definition of continuous symmetries of the action to allow for a boundary term for small s as follows:

$$\begin{aligned} I[\mathbf{q}^*] &= \int_{t_A}^{t_B} L(\mathbf{q}^*(t), \dot{\mathbf{q}}^*(t), t) dt = \int_{t_A}^{t_B} L(\mathbf{q}(t), \dot{\mathbf{q}}(t), t) dt + [sK(\mathbf{q}, \dot{\mathbf{q}}, t)]_{t_A}^{t_B} \\ &= I[\mathbf{q}] + [sK(\mathbf{q}, \dot{\mathbf{q}}, t)]_{t_A}^{t_B} \end{aligned}$$

Going once again through the proof of Noether's theorem, we obtain the conserved quantity

$$\sum_i \frac{\partial L}{\partial \dot{q}_i} \left(\frac{\partial Q_i}{\partial s} \right)_{s=0} - K$$

Example

Take

$$L = \sum_{i=1}^N \frac{1}{2} m_i \dot{\mathbf{x}}_i^2 - V(\mathbf{x}_1, \dots, \mathbf{x}_N)$$

where V depends only on the pairwise differences. Define a Galilean boost $\mathbf{x}^* = \mathbf{x} - \mathbf{v}t$. Set $\mathbf{v} = \hat{\mathbf{n}}s$, so we obtain

$$\mathbf{Q}(\mathbf{q}) = (\mathbf{x}_1 - \hat{\mathbf{n}}st, \dots, \mathbf{x}_N - \hat{\mathbf{n}}st)$$

Then

$$\begin{aligned} L(\mathbf{x}^*, \dot{\mathbf{x}}^*, t) &= \sum_{i=1}^N \frac{1}{2} m_i (\dot{\mathbf{x}}_i - \hat{\mathbf{n}}s)^2 - V(\mathbf{x}_1^*, \dots, \mathbf{x}_N^*) \\ &= \sum_{i=1}^N \frac{1}{2} m_i \dot{\mathbf{x}}_i^2 - V(\mathbf{x}_1, \dots, \mathbf{x}_N) - s\hat{\mathbf{n}} \cdot \sum m_i \dot{\mathbf{x}}_i + O(s^2) \\ &= L(\mathbf{x}, \dot{\mathbf{x}}, t) - \frac{d}{dt} \left(s\hat{\mathbf{n}} \cdot \sum_i m_i \mathbf{x}_i \right) \end{aligned}$$

so we have the scenario from above with $K = -\hat{\mathbf{n}} \cdot \sum_i m_i \mathbf{x}_i = -\hat{\mathbf{n}} \cdot (M\mathbf{R})$, where M is the total mass and $\mathbf{R}$ the centre of mass. Hence, we have the conserved quantity

$$\begin{aligned} \sum_i \frac{\partial L}{\partial \dot{q}_i} \left(\frac{\partial Q_i}{\partial s} \right)_{s=0} - K &= \sum_i \mathbf{p}_i \cdot (-\hat{\mathbf{n}}t) + M\hat{\mathbf{n}} \cdot \mathbf{R} \\ &= \hat{\mathbf{n}} \cdot (M\mathbf{R} - \mathbf{P}t) \end{aligned}$$

so, since $\hat{\mathbf{n}}$ was arbitrary, $M\mathbf{R} - \mathbf{P}t$ is conserved.

Of course, there is an analogous statement of Noether's theorem for symmetry of the action under transformation of the time coordinate. Define a new time coordinate $t^* = T(t)$, for invertible T , so

$$\mathbf{q}^*(t^*) = \mathbf{q}(t) = \mathbf{q}(T^{-1}(t^*))$$

This is a symmetry of the action if

$$\int_{t_A^*}^{t_B^*} L\left(\mathbf{q}^*(t^*), \frac{d\mathbf{q}^*}{dt^*}(t^*), t^*\right) dt^* = \int_{t_A}^{t_B} L(\mathbf{q}(t), \dot{\mathbf{q}}(t), t) dt \quad (\dagger)$$

for all $\mathbf{q}(t)$, t_A and t_B . Let us consider a one-parameter family of these transformations, $T(t, s)$, with

$$t^* = t + s \left(\frac{\partial T}{\partial s} \right)_{s=0} + O(s^2)$$

Theorem 9.4 (Noether's theorem, 2)

If there exists a 1-parameter family of symmetries of the action of this type, then any solution of the Euler-Lagrange equations has a conserved quantity given by

$$\left(L - \sum_i \dot{q}_i \frac{\partial L}{\partial \dot{q}_i} \right) \left(\frac{\partial T}{\partial s} \right)_{s=0}$$

Proof. Observe

$$\frac{dt^*}{dt} = 1 + s \frac{d}{dt} \left(\frac{\partial T}{\partial s} \right)_{s=0} + O(s^2)$$

Let I^* , I be the LHS and RHS of (†), respectively. We start by changing the variable of integration in I^* from t^* to t , obtaining

$$I^* = \int_{t_A}^{t_B} L \left(\mathbf{q}(t), \frac{dt}{dt^*} \frac{d}{dt} \mathbf{q}(t), T(t) \right) \frac{dt^*}{dt} dt$$

Taking the reciprocal of $\frac{dt^*}{dt}$ and expanding using the binomial theorem, we obtain

$$\frac{dt}{dt^*} = 1 - s \frac{d}{dt} \left(\frac{\partial T}{\partial s} \right)_{s=0} + O(s^2)$$

so that

$$\begin{aligned} I^* &= \int_{t_A}^{t_B} L \left(\mathbf{q}(t), \dot{\mathbf{q}}(t) - s \frac{d}{dt} \left(\frac{\partial T}{\partial s} \right)_{s=0} \dot{\mathbf{q}}(t), t + s \left(\frac{\partial T}{\partial s} \right)_{s=0} \right) \left(1 + s \frac{d}{dt} \left(\frac{\partial T}{\partial s} \right)_{s=0} \right) dt \\ &= \int_{t_A}^{t_B} L(\mathbf{q}(t), \dot{\mathbf{q}}(t), t) - s \frac{d}{dt} \left(\frac{\partial T}{\partial s} \right)_{s=0} \dot{q}_i \frac{\partial L}{\partial \dot{q}_i} + s \left(\frac{\partial T}{\partial s} \right)_{s=0} \frac{\partial L}{\partial t} + s L \frac{d}{dt} \left(\frac{\partial T}{\partial s} \right)_{s=0} dt \\ &= I + \underbrace{\int_{t_A}^{t_B} \frac{d}{dt} \left(\frac{\partial T}{\partial s} \right)_{s=0} \left(-s \dot{q}_i \frac{\partial L}{\partial \dot{q}_i} + s L \right)}_{\text{integrate by parts}} + s \left(\frac{\partial T}{\partial s} \right)_{s=0} \frac{\partial L}{\partial t} dt \\ &= I + \left[s \left(\frac{\partial T}{\partial s} \right)_{s=0} \left(L - \dot{q}_i \frac{\partial L}{\partial \dot{q}_i} \right) \right]_{t_A}^{t_B} + \underbrace{\int_{t_A}^{t_B} s \left(\frac{\partial T}{\partial s} \right)_{s=0} \left(-\frac{d}{dt} \left(L - \dot{q}_i \frac{\partial L}{\partial \dot{q}_i} \right) + \frac{\partial L}{\partial t} \right) dt}_{0 \text{ by E-L}} \\ &= I + \left[s \left(\frac{\partial T}{\partial s} \right)_{s=0} \left(L - \dot{q}_i \frac{\partial L}{\partial \dot{q}_i} \right) \right]_{t_A}^{t_B} \end{aligned}$$

Hence, since $I^* = I$,

$$s \left(\frac{\partial T}{\partial s} \right)_{s=0} \left(L - \dot{q}_i \frac{\partial L}{\partial \dot{q}_i} \right) = \text{const.}$$

which gives the conserved quantity as required. ■

Example

Suppose L has no explicit t -dependence, and define $T(t, s) = t + s$. We have

$$\mathbf{q}^*(t^*) = \mathbf{q}(t) = \mathbf{q}(t^* - s) \implies \frac{\partial \mathbf{q}^*}{\partial t^*} = \dot{\mathbf{q}}(t^* - s)$$

so

$$\begin{aligned} \int_{t_A^*}^{t_B^*} L\left(\mathbf{q}^*(t^*), \frac{d\mathbf{q}^*}{dt^*}(t^*)\right) dt^* &= \int_{t_A+s}^{t_B+s} L(\mathbf{q}(t^*-s), \dot{\mathbf{q}}(t^*-s)) dt^* \\ &= \int_{t_A}^{t_B} L(\mathbf{q}(t), \dot{\mathbf{q}}(t)) dt \end{aligned}$$

by a simple substitution. Note that this only worked since the Lagrangian was independent of t . Hence, we have the conserved quantity

$$\underbrace{\left(\frac{\partial T}{\partial s}\right)}_{s=0} \left(L - \dot{q}_i \frac{\partial L}{\partial \dot{q}_i}\right) = -E$$

Hence, we have recovered conservation of energy as a consequence of invariance under time translation.

§10 PDEs from variational principles

Consider functionals of functions

$$\mathbf{y} : \mathbb{R}^m \rightarrow \mathbb{R}^n$$

of the form

$$F[\mathbf{y}] = \int_V f(\mathbf{y}, \nabla \mathbf{y}, x_1, \dots, x_m) dx_1 \dots dx_m$$

where $V \subseteq \mathbb{R}^m$, and $\nabla \mathbf{y} = \left(\frac{\partial \mathbf{y}}{\partial x_1}, \dots, \frac{\partial \mathbf{y}}{\partial x_m}\right)$ is an m by n matrix. Varying $\mathbf{y}$ and integrating by parts using the divergence theorem, we obtain a generalisation of the Euler-Lagrange equations, provided boundary terms vanish as before.

§10.1 Minimal surfaces

Consider surfaces spanning a closed curve C in $\mathbb{R}^3$. Which has the minimal area? We will restrict to surfaces which are the graphs of functions of x and y , i.e. $z = h(x, y)$, where C has equation $z = h(x, y)$ for $x, y \in \partial D$ – so h is fixed on ∂D . Let us parameterise using (x, y) to obtain

$$\mathbf{x}(x, y) = (x, y, h(x, y))$$

giving

$$d\mathbf{S} = \frac{\partial \mathbf{x}}{\partial x} \times \frac{\partial \mathbf{x}}{\partial y} dx dy = \begin{pmatrix} 1 \\ 0 \\ h_x \end{pmatrix} \times \begin{pmatrix} 0 \\ 1 \\ h_y \end{pmatrix} dx dy = \begin{pmatrix} -h_x \\ -h_y \\ 1 \end{pmatrix} dx dy$$

$$\implies dS = \sqrt{1 + h_x^2 + h_y^2} dx dy$$

so the surface has total area

$$A[h] = \int_D \sqrt{1 + h_x^2 + h_y^2} dx dy$$

Now, expanding to first order,

$$\begin{aligned} \delta A &= \int_D \frac{h_x(\delta h)_x + h_y(\delta h)_y}{\sqrt{1 + h_x^2 + h_y^2}} dx dy \\ &= \int_D \left(\left(\frac{h_x(\delta h)_x}{\sqrt{\dots}} \right)_x + \left(\frac{h_y(\delta h)_y}{\sqrt{\dots}} \right)_y - \left(\left(\frac{h_x}{\sqrt{\dots}} \right)_x + \left(\frac{h_y}{\sqrt{\dots}} \right)_y \right) \delta h \right) dx dy \end{aligned}$$

Applying Green's theorem to the first term produces a boundary term which is zero on ∂D . Hence, A is extremised only if

$$\begin{aligned} 0 &= \left(\frac{h_x}{\sqrt{\dots}} \right)_x + \left(\frac{h_y}{\sqrt{\dots}} \right)_y \\ &= (1 + h_y^2)h_{xx} + (1 + h_x^2)h_{yy} - 2h_x h_y h_{xy} = 0 \end{aligned} \quad (*)$$

which is called the *minimal surface equation*. This is a second-order non-linear PDE for $h(x, y)$, so is hard to solve. Notably, however, if $|h_x| \ll 1$ and $|h_y| \ll 1$, however, this is approximately $h_{xx} + h_{yy} = 0$, which is Laplace's equation.

Anyhow, note that, if all second derivatives are zero, the equation is certainly satisfied. Hence

$$h = Ax + By + C$$

that is, a plane, is a solution. But this only satisfies the boundary conditions if the curve C is planar.

Another approach would be to look for surfaces of revolution about the z -axis, i.e. $h(x, y) = z(r)$, where $r = \sqrt{x^2 + y^2}$. Plugging this into (*), we obtain

$$rz'' + z' + z'^3 = 0$$

Disturbingly, this is still hard to solve. A better approach would be to plug this back into the original area functional, to obtain

$$A[z] = 2\pi \int_D r \sqrt{1 + z'^2} dr$$

Indeed, this equation has a conserved quantity, and we obtain

$$r = r_0 \cosh \left(\frac{z - z_0}{r_0} \right)$$

which is a catenoid.

§10.2 Vibrating string

A string with tension T and mass per unit length ρ has

$$\begin{aligned}\text{KE} &= \frac{1}{2}\rho \int_0^a \dot{y}(t, x)^2 dx \\ \text{PE} &= \frac{1}{2}T \int_0^a y'(t, x)^2 dx\end{aligned}$$

where dot and prime denote $\frac{\partial}{\partial t}$ and $\frac{\partial}{\partial x}$ respectively. Hence the action is given by

$$I[y] = \frac{1}{2} \int_{t_0}^{t_1} \int_0^a (\rho \dot{y}^2 - T y'^2) dx dt$$

which, upon variation, gives

$$\begin{aligned}\delta I &= \int_{t_0}^{t_1} \int_0^a (\rho \dot{y} \delta \dot{y} - T y' \delta y') dx dt \\ &= \int_{t_0}^{t_1} \int_0^a (\partial_t(\rho \dot{y} \delta y) - \partial_x(T y' \delta y) + (-\rho \ddot{y} + T y'')) \delta y dx dt\end{aligned}$$

Integrating the derivatives away to boundary terms, we obtain

$$\ddot{y} - v^2 y'' = 0$$

where $v = \sqrt{T/\rho}$ – the wave equation! The general solution is given by

$$y(t, x) = f(x - vt) + g(x + vt)$$

i.e., a superposition of a right and left-moving wave.

§10.3 Maxwell's equations

Consider electric and magnetic fields $\mathbf{E}(t, \mathbf{x})$ and $\mathbf{B}(t, \mathbf{x})$ – then Maxwell's equations are given by

$$\begin{aligned}\nabla \cdot \mathbf{B} &= 0 \\ \nabla \times \mathbf{E} &= -\partial_t \mathbf{B}\end{aligned}$$

Hence, since $\mathbf{B}$ is solenoidal, $\mathbf{B} = \nabla \times \mathbf{A}$ for some $\mathbf{A}(t, \mathbf{x})$, giving

$$\nabla \times (\mathbf{E} + \partial_t \mathbf{A}) = 0$$

Now, this expression is conservative, so $\mathbf{E} = -\nabla \phi - \partial_t \mathbf{A}$ for some scalar potential ϕ . The action is then a functional of $\mathbf{A}$ and ϕ – it is

$$I[\mathbf{A}, \phi] = \int \left(\frac{1}{2} \varepsilon_0 \mathbf{E}^2 - \frac{1}{2\mu_0} \mathbf{B}^2 + \mathbf{A} \cdot \mathbf{j} - \phi \rho \right) d^3x dt$$

where $\mathbf{j}(t, \mathbf{x})$ is the *current density* and $\rho(t, \mathbf{x})$ is the *charge density*. Varying $\mathbf{A}$ and ϕ , we obtain

$$\delta I = \int \left(\varepsilon_0 \mathbf{E} \cdot (-\nabla \delta \phi - \partial_t \delta \mathbf{A}) - \frac{1}{\mu_0} \mathbf{B} \cdot (\nabla \times \delta \mathbf{A}) + \delta \mathbf{A} \cdot \mathbf{j} - \delta \phi \rho \right) d^3x dt$$

$$\begin{aligned}
&= \int (\varepsilon_0 (-\nabla \cdot (\mathbf{E}\delta\phi) + \nabla \cdot \mathbf{E}\delta\phi - \partial_t(\mathbf{E} \cdot \delta\mathbf{A}) + (\partial_t\mathbf{E}) \cdot \delta\mathbf{A}) - \\
&\quad \frac{1}{\mu_0} (\partial_j(\varepsilon_{ijk}B_i\delta A_k) - (\varepsilon_{ijk}\delta_j B_i)\delta A_k) + \delta\mathbf{a} \cdot \mathbf{j} - \delta\phi\rho) d^3x dt \\
&= \text{boundary terms} + \int \left((\varepsilon_0 \nabla \cdot \mathbf{E} - \rho)\delta\phi + \left(-\frac{1}{\mu_0} \nabla \times \mathbf{B} + \varepsilon_0 \partial_t \mathbf{E} + \mathbf{j} \right) \cdot \delta\mathbf{A} \right) d^3x dt
\end{aligned}$$

Hence, making $\delta I = 0$ for arbitrary $\delta\phi, \delta\mathbf{A}$, we obtain

$$\begin{aligned}
\nabla \cdot \mathbf{E} &= \frac{\rho}{\varepsilon_0} \\
\nabla \times \mathbf{B} &= \mu_0 \left(\mathbf{j} + \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right)
\end{aligned}$$

§11 The second variation

Recall that, for a function, we write

$$f(\mathbf{x} + \varepsilon \boldsymbol{\xi}) = f(\mathbf{x}) + \varepsilon \boldsymbol{\xi} \cdot \nabla f(\mathbf{x}) + \frac{1}{2} \varepsilon^2 \xi_i \xi_j H_{ij}(\mathbf{x}) + O(\varepsilon^3)$$

so that a stationary point $\mathbf{a}$ is a local minimum if $H_{ij}(\mathbf{a})$ is positive-definite, i.e. $\xi_i \xi_j H_{ij}(\mathbf{a}) > 0$ for any $\boldsymbol{\xi} \neq \mathbf{0}$, or, equivalently, $H_{ij}(\mathbf{a})$ has all positive eigenvalues.

Consider, now, a functional

$$F[y] = \int_{\alpha}^{\beta} f(y, y', x) dx$$

with $y \in \mathcal{C} := \{y \in C^2, y : [\alpha, \beta] \rightarrow \mathbb{R}, y(\alpha) = a, y(\beta) = b\}$. Consider $y + \varepsilon \xi$, where we must have $\xi(\alpha) = \xi(\beta) = 0$ to retain boundary conditions. For small ε , we have

$$\begin{aligned}
f(y + \varepsilon \xi, y' + \varepsilon \xi', x) &= f(y, y', x) + \varepsilon \xi \left(\frac{\partial f}{\partial y} - \frac{d}{dx} \frac{\partial f}{\partial y'} \right) + \varepsilon \frac{d}{dx} \left(\xi \frac{\partial f}{\partial y'} \right) \\
&\quad + \frac{1}{2} \varepsilon^2 \left(\xi^2 \frac{\partial^2 f}{\partial y^2} + 2\xi \xi' \frac{\partial^2 f}{\partial y \partial y'} + \xi'^2 \frac{\partial^2 f}{\partial y'^2} \right) + O(\varepsilon^3)
\end{aligned}$$

so

$$F[y + \varepsilon \xi] - F[y] = \varepsilon \int_{\alpha}^{\beta} \xi(x) \frac{\delta F[y]}{\delta y(x)} dx + \varepsilon^2 \delta^2 F[y, \xi] + O(\varepsilon^3)$$

where the boundary terms have vanished due to the constraint on ξ , and

$$\delta^2 F[y, \xi] := \frac{1}{2} \int_{\alpha}^{\beta} \left(\xi^2 \frac{\partial^2 f}{\partial y^2} + 2\xi \xi' \frac{\partial^2 f}{\partial y \partial y'} + \xi'^2 \frac{\partial^2 f}{\partial y'^2} \right) dx$$

Note that $2\xi \xi' = (\xi^2)'$, so we integrate this term by parts to obtain

$$\delta^2 F[y, \xi] = \frac{1}{2} \int_{\alpha}^{\beta} \left(\xi^2 \left(\frac{\partial^2 f}{\partial y^2} - \frac{d}{dx} \frac{\partial^2 f}{\partial y \partial y'} \right) + \xi'^2 \frac{\partial^2 f}{\partial y'^2} \right) dx$$

Recall that, in the case of functions, if $H(\mathbf{x})$ is positive definite everywhere, then $f(\mathbf{x})$ is convex, so any stationary point is a global minimum. Similarly, for functionals, if $\delta^2 F[y, \xi] \geq 0$ for all $y \in \mathcal{C}$ and allowed ξ , then $y = y_0$ is a global minimum of F so long as y_0 satisfies the Euler-Lagrange equations.

Remark. Note that $\mathcal{C}$ is a convex set, since it is closed under linear combinations.

Example (Geodesics in the Euclidean plane)

Recall that

$$F[y] = \int_{\alpha}^{\beta} \sqrt{1 + y'^2} dx$$

is the length of a path in the Euclidean plane. Observe that the first two terms in $\delta^2 F[y, \xi]$ are both zero, since F has no explicit dependence on y , so we obtain

$$\delta^2 F[y, \xi] = \frac{1}{2} \int_{\alpha}^{\beta} \xi'^2 (1 + y'^2)^{3/2}$$

which is clearly positive for all y, ξ . Hence any solution of the Euler-Lagrange equation is a global minimum.

However, we can make weaker statements. Indeed, if $\delta^2 F[y_0, \xi]$ is positive for all non-trivial allowed ξ and a fixed y_0 , where y satisfies the Euler-Lagrange equations, then y_0 is a local minimum. Conversely, if $\delta^2 F[y_0, \xi] < 0$ for some allowed ξ , then y_0 cannot be a local minimum.

Example (Geodesics on a sphere)

A geodesic between points A, B on a sphere is a great circle through these points. However, there is some subtlety, since there is a minor and major arc, one of which is longer. Evaluating $\delta^2 F$ on each arc will give that the minor arc is the local minimum, while the major arc has $\delta^2 F < 0$ for some ξ .

Proposition 11.1 (Legendre condition)

If $y \in \mathcal{C}$ satisfies the Euler-Lagrange equations, a *necessary* criterion for it to be a local minimum is that

$$\left(\frac{\partial^2 f}{\partial y'^2} \right)_{y=y_0} \geq 0$$

for every $x \in [\alpha, \beta]$.

Proof. Observe that, if

$$\left(\frac{\partial^2 f}{\partial y'^2} \right)_{y=y_0} < 0$$

at some $x = x_0 \in [\alpha, \beta]$, then we may choose ξ such that it is non-zero only near x_0 , and ξ' is large with ξ small. This allows us to make $\delta^2 F[x, \xi]$ negative, as required. ■

Example

Take

$$F[y] = \int_{-1}^1 x \sqrt{1 + y'^2} \, dx$$

so

$$\frac{\partial^2 f}{\partial y'^2} = x(1 + y'^2)^{-3/2}$$

which violates the Legendre condition for $x < 0$. Hence this functional has no local minima.

Let us introduce some new notation – given y_0 satisfying the Euler-Lagrange equations, write

$$\delta^2 F[y_0, \xi] = \frac{1}{2} \int_{\alpha}^{\beta} (\rho(x) \xi'^2 + \sigma(x) \xi^2) \, dx \quad (*)$$

where

$$\rho = \left(\frac{\partial^2 f}{\partial y'^2} \right)_{y=y_0} \quad \sigma = \left(\frac{\partial^2 f}{\partial y^2} - \frac{d}{dx} \frac{\partial^2 f}{\partial y \partial y'} \right)_{y=y_0}$$

Then $\rho \geq 0$ is a necessary condition for y_0 to be a local minimum, while $\rho > 0$, $\sigma \geq 0$ is a sufficient condition.

Proof. Clearly $\delta^2 F \geq 0$ for such ρ , σ . If $\delta^2 F = 0$, then ξ' vanishes. Then ξ is a constant, but, since ξ vanishes at the endpoints, it must vanish everywhere, so it is trivial. ■

Example (Brachistochrone)

Recall that

$$T[y] = \int_0^{x_B} \sqrt{\frac{1 + y'^2}{-y}}$$

(we add the condition $y < 0$). Then

$$\begin{aligned} \frac{\partial f}{\partial y'} &= \frac{y'}{\sqrt{-y(1 + y'^2)}} & \frac{\partial f}{\partial y} &= -\frac{f}{2y} \\ \Rightarrow \rho &= \frac{\partial^2 f}{\partial y'^2} = \frac{1}{\sqrt{-y(1 + y'^2)^3}} > 0 \\ \sigma &= \frac{\partial^2 f}{\partial y^2} - \frac{d}{dx} \left(-\frac{1}{2y} \frac{\partial f}{\partial y'} \right) = \frac{\partial^2 f}{\partial y^2} - \frac{y'}{2y^2} \frac{\partial f}{\partial y'} + \frac{1}{2y} \frac{\partial f}{\partial y} \\ &= \frac{3f}{4y^2} - \frac{y'^2}{2y^2 \sqrt{-y(1 + y'^2)}} - \frac{f}{4y^2} = \frac{1}{2\sqrt{-y(1 + y'^2)}} > 0 \end{aligned}$$

where we have used the Euler-Lagrange equation in the evaluation of σ . Hence the cycloid is a local minimum, as required.

Observe integrating (*) by parts gives

$$\delta^2 F[y_0, \xi] = \frac{1}{2} \int_{\alpha}^{\beta} \xi(x) \mathcal{L}\xi(x) dx + \underbrace{\frac{1}{2} [\rho \xi \xi']_{\alpha}^{\beta}}_0 \quad (**)$$

where

$$\mathcal{L}\xi = -\frac{d}{dx} \left(\rho \frac{d\xi}{dx} \right) + \sigma \xi$$

as in [section 7.1](#).

Proposition 11.2 (Second variation and eigenvalues of the Sturm-Liouville problem)

Consider the following Sturm-Liouville problem:

$$\begin{aligned} \mathcal{L}\eta &= \lambda w(x) \eta \\ \eta(\alpha) &= \eta(\beta) = 0 \end{aligned}$$

where $w(x) > 0$ on (α, β) . y_0 is a local minimum if and only if all eigenvalues admitted by this problem are positive.

Proof. $\Rightarrow$ Suppose there is a negative eigenvalue λ with corresponding eigenfunction η . Set $\xi = \eta$ in (**), so we obtain

$$\delta^2 F[y_0, \eta] = \frac{1}{2} \int_{\alpha}^{\beta} \eta \mathcal{L}\eta dx = \frac{\lambda}{2} \int_{\alpha}^{\beta} w \eta^2 dx < 0$$

which suffices.

$\Leftarrow$ Let λ_n and $y_n(x)$ be the n^{th} eigenvalue and eigenfunction respectively. Recall from 1B Methods that $\{y_n\}$ form a complete set for the space of allowed η , so that every η is expressible as

$$\eta(x) = \sum_n a_n y_n(x)$$

for some a_n , and furthermore we may choose y_n to be orthogonal with respect to the weight function w , so

$$\int_{\alpha}^{\beta} w y_m y_n dx = 0$$

for $m \neq n$. Now, (**) gives

$$\begin{aligned} \delta^2 F[y_0, \eta] &= \frac{1}{2} \sum_{m,n} a_m a_n \lambda_n \int_{\alpha}^{\beta} w y_m y_n dx \\ &= \frac{1}{2} \sum_m \lambda_m a_m^2 \int_{\alpha}^{\beta} w y_m^2 dx \geq 0 \end{aligned}$$

by orthogonality. But observe this vanishes only if $a_m = 0$ for all m , in which case $\eta \equiv 0$. Hence, if $\eta \not\equiv 0$, the second variation is strictly positive as required. ■

Example

Take

$$F[y] = \frac{1}{2} \int_0^a (y'^2 - y^2) dx$$

where $y(0) = y(a)$. The Euler-Lagrange equation is

$$-y - \frac{d}{dx}(y') = 0 \iff y'' + y = 0$$

Suppose we have some solution y_0 , and expand to obtain

$$\delta^2 F = \frac{1}{2} \int_0^a (\xi'^2 - \xi^2) dx$$

for which $\rho = 1$, $\sigma = -1$. Hence, the associated Sturm-Liouville problem is

$$-\frac{d^2\eta}{dx^2} - \eta = \lambda w\eta$$

where $\eta(0) = \eta(a) = 0$. Let us select $w(x) = 1$, and we obtain

$$\begin{aligned} 0 &= \eta'' + (\lambda + 1)\eta \\ \implies \eta &= A \sin(x\sqrt{\lambda + 1}) + B \cos(x\sqrt{\lambda + 1}) \end{aligned}$$

Now, applying the boundary conditions, we have that $\eta(0) = 0$ implies $B = 0$, but furthermore that $\eta(a) = A \sin(a\sqrt{\lambda + 1}) = 0$. Hence, a non-trivial solution must have $a\sqrt{\lambda + 1} = n\pi$, giving

$$\eta = A \sin\left(\frac{n\pi x}{a}\right)$$

and we may without loss of generality take $n \in \mathbb{N}$. The eigenvalues, then, are

$$\lambda_n = \frac{n^2\pi^2}{a^2} - 1$$

Since these are increasing in n , they are all positive if and only if $\lambda_1 > 0$, that is $a < \pi$. Otherwise, if $a > \pi$, we do not have a local minimum.

§12 The Jacobian condition

The Jacobian condition is another, slightly less sharp, method for determining whether solutions to the Euler-Lagrange equations are local minima. Given a $y_0 \in \mathcal{C}$, observe

$$0 = \int_\alpha^\beta (\phi\xi^2)' dx = \int_\alpha^\beta 2\phi\xi\xi' + \phi'\xi^2 dx$$

for any ϕ . Adding this to (*), we obtain

$$\delta^2 F[y_0, \xi] = \frac{1}{2} \int_\alpha^\beta (\rho\xi'^2 + 2\phi\xi\xi' + (\sigma + \phi')\xi^2) dx$$

Assume $\rho > 0$ for $\alpha \leq x \leq \beta$, and we complete the square to obtain

$$\delta^2 F[y_0, \xi] = \frac{1}{2} \int_{\alpha}^{\beta} \left(\rho \left(\xi' + \frac{\phi}{\rho} \xi \right)^2 + \left(\sigma + \phi' - \frac{\phi^2}{\rho} \right) \xi^2 \right) dx$$

Now, the first term is nice and positive, so we would like to choose ϕ to make the second term vanish, i.e. satisfying the Ricatti equation

$$\phi' + \sigma = \frac{\phi^2}{\rho} \quad (**)$$

If we can pull this off, then $\delta^2 F[y_0, \xi] \geq 0$ if and only if $\xi' + \frac{\phi}{\rho} \xi$ vanishes, in which case

$$\xi(x) = C \exp \left(- \int_{\alpha}^x \frac{\phi(s)}{\rho(s)} ds \right)$$

However, since $\xi(\alpha) = 0$, $C = 0$, so $\xi(x) \equiv 0$. So $\delta^2 F[y_0, \xi]$ is strictly positive for all non-trivial allowed ξ . We conclude that y_0 is a local minimum if there exists a solution to $(**)$ on the interval $[\alpha, \beta]$.

This is a priori hard to solve, however we can convert to a linear second order ODE by means of the substitution $\phi = -\rho \frac{u'}{u}$ for some $u \neq 0$. We obtain

$$\begin{aligned} \rho \frac{u'^2}{u^2} &= \sigma - \left(\rho \frac{u'}{u} \right)' \\ &= \sigma - \frac{1}{u} (\rho u')' + \rho \frac{u'^2}{u^2} \end{aligned}$$

and the non-linear terms cancel to give the *Jacobi accessory equation*

$$-(\rho u')' + \sigma u = 0$$

While this can always be solved, the solution may not be everywhere non-zero.

Example

Let us consider again

$$F[y] = \frac{1}{2} \int_0^a (y'^2 - y^2) dx$$

which gives $\rho = 1$, $\sigma = -1$. The Jacobi accessory equation is

$$-u'' - u = 0 \implies u = A \sin(x - x_0)$$

Of course, these solutions oscillate with period 2π , and have roots on any interval of size at least π – so we can find a solution with $u \neq 0$ on $[0, a]$ if and only if $a < \pi$. Hence we can determine that a solution y_0 is a local minimum if $a < \pi$, but not the converse as we had with Sturm-Liouville – we cannot show that the solutions for $a > \pi$ fail to be local minima.